

A historical introduction to the Riemann Hypothesis

Z M Tanver Mahmud Anindya¹

1 Introduction

If you were asked how hard it could be to become a millionaire, then you might say that it is incredibly challenging, as it requires talent, luck and hard work. However, in the realm of mathematics, there exists a single problem that is so elegant and profound that solving this problem will make someone a millionaire: the Riemann Hypothesis. Proposed by the German mathematician Bernhard Riemann [8] in 1859, this mysterious conjecture has haunted mathematicians ever since. This hypothesis might hold the key to the mysteries of prime numbers, along with number theory, complex analysis and even the deep secrets of physics.

This article provides a historical introduction to the Riemann Hypothesis based primarily on accessible sources such as Wikipedia [14] and Encyclopedia Britannica [5]. To better visualize and understand complex concepts, explanations from educational YouTube channels including Quanta Magazine [7], Singing Banana [10], Numberphile [6] and 3Blue1Brown [1] have been invaluable. Since the original 1859 paper [8] by Riemann is in German and therefore less accessible these days, this article synthesizes these sources for a broader audience.



Figure 1: Leonhard Euler [2]

¹Z M Tanver Mahmud Anindya is a 2nd-year undergraduate student of Mechanical Engineering at Khulna University of Engineering Technology, Bangladesh

Back in the early 18th century, mathematicians were diving deeply into infinite series. The great mathematician Leonhard Euler was among these pioneers. In 1734, he solved a famous problem, called the *Basel Problem*, by showing that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots = \frac{\pi^2}{6}.$$

Later, he extended this series for higher powers by showing that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945},$$

and so on. By these identities, he found an astounding connection between infinite series and geometry. Euler then generalized these series as what is now known as the *zeta function*. It looks like this:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

where s is any real number greater than 1. Euler found that the series converges for any value s greater than 1. But for other values of s , the series might diverge. For example, if $s = 1$, then

$$\zeta(1) = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots = \infty$$

which is a divergent harmonic series. Euler did not stop here. He showed that, by multiplying the function by $(1 - \frac{1}{p^s})$ for each prime p , every term gets cancelled out. Euler's product form of the the zeta function is therefore

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1}.$$

This formula shows that the zeta function $\zeta(s)$ is connected with prime numbers.

Talking about the primes, at the close of the 18th century, many mathematicians were investigating prime numbers. Based on gathered data for primes up to three million, Adrien-Marie Legendre, Carl Fredrich Gauss and others noticed that the frequency of primes decreases in a predictable manner. In particular, they found that the number of primes less than any number x is close to $\frac{x}{\ln x}$; that is, $\pi(x) \approx \frac{x}{\ln x}$, where $\pi(x)$ indicates the number of primes less than x . Later, in 1838, Peter Gustav Lejeune Dirichlet developed a more accurate expression for the distribution of prime numbers:

$$\pi(x) \approx \text{Li}(x) = \int_2^x \frac{dt}{\ln t}.$$

This formula gives a much better estimation than the previous one. The reason why the logarithm appears in prime distribution is straightforward. The density of prime

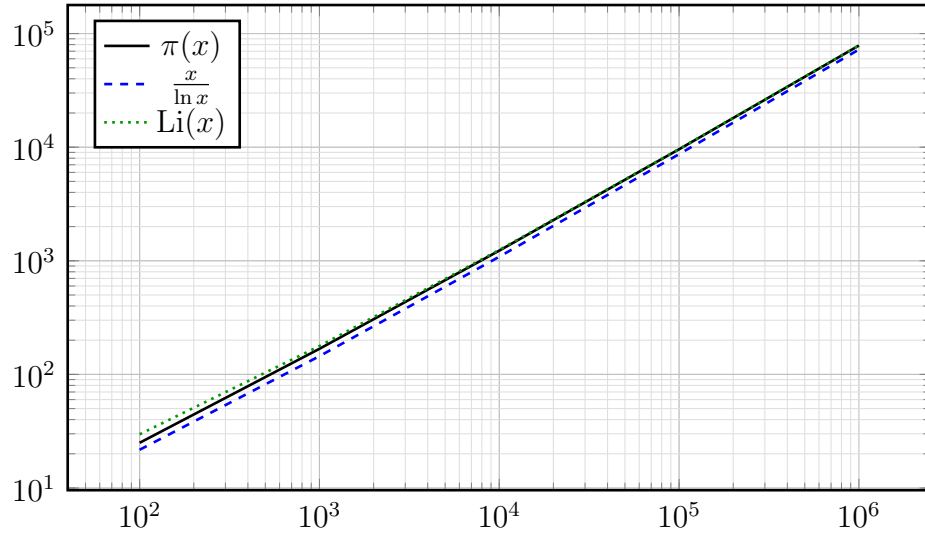


Figure 2: Comparison of actual prime count $\pi(x)$ with $\frac{x}{\ln x}$ and $\text{Li}(x)$.

numbers decreases as the number increases. So, after plotting the data up to three million, these mathematicians saw that a nearly smooth curve appeared as the number increased. To get the total number of prime numbers, their decreasing frequency or density must be considered. That's why the logarithmic integral appears in the prime distribution.

We can also see it from Euler's product formula. Taking natural logarithm on both sides, the series simplifies which denotes the underlying connection between logarithm and prime numbers.



Figure 3: Bernhard Riemann [11]

A little later, Bernhard Riemann published his groundbreaking paper [8] in 1859, upending centuries of studies on prime numbers. This was his only paper on number

theory. The problem with Euler's zeta function is that it is only valid for $s > 1$, which ignores the vast territory of negative and complex numbers. Riemann extended the horizon of the zeta function by letting the function be valid for any complex variable of the form where $s \in \mathbb{C}$ and $s \in \mathbb{R}$, except $s = 1$, where the function diverges. Riemann used so-called *analytic continuation* to discover a functional equation that extended the zeta function to all complex and negative values except $s = 1$. Analytic continuation is a method of extending the domain of any function to such territories where it usually doesn't apply, while preserving the core essence of the original function. In particular, Riemann proved the following equation:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

This may look complicated, as it includes the trigonometric sine function and the gamma function $\Gamma(x)$. However, it highlights an important part of the function: symmetry. If we know the value of the zeta function for any s , then we can determine the value of the function at $1-s$. This creates a mirror image on the complex plane which is symmetrical across the line where the real part of s is $\Re(s) = \frac{1}{2}$. This line is called the *critical line*. The region between $\Re(s) = 0$ and $\Re(s) = 1$ is called the *critical strip*.

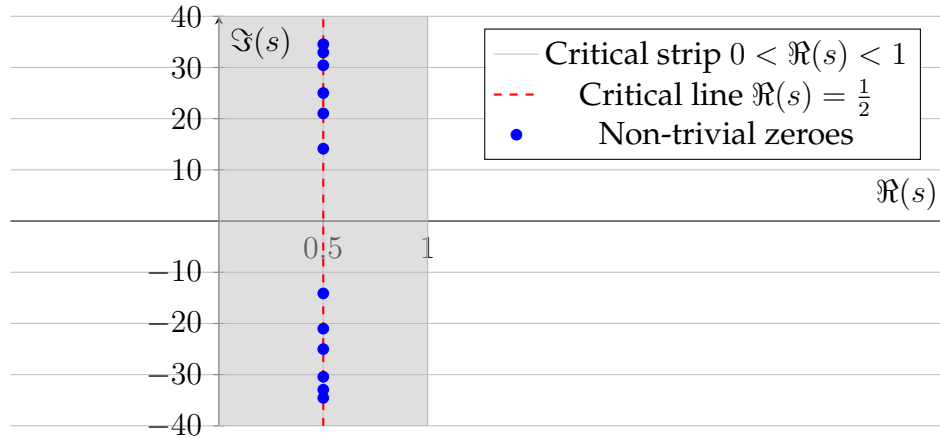


Figure 4: Symmetry of $\zeta(s)$ on the critical line $\Re(s) = \frac{1}{2}$

To explain the symmetry more clearly, Riemann introduced a modified version of the zeta function, denoted by $\xi(s)$, which has no pole at $s = 1$.

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

The magic of this function is it satisfies the relation $\xi(s) = \xi(1-s)$, making it perfectly symmetrical across the critical line.

Riemann introduced the zeroes of the zeta function. Some zeroes are easy to find by plugging some real values of s . These are called trivial zeroes. Other zeroes lie on the complex plane which are called non-trivial zeroes. These zeroes are found in

the critical strip. But Riemann boldly hypothesized that all the non-trivial zeroes lie exactly on the critical line where $\Re(s) = \frac{1}{2}$. This is known as the *Riemann Hypothesis*.

Now why do these zeroes matter so much? Remember the approximation $\pi(x) \approx \frac{x}{\ln x}$? That theorem estimates the number of primes up to any given number but there is always small gap between the actual count of prime and the theoretical count. Riemann showed that these non-trivial zeroes control the error. If the zeroes lie on the critical line, then the error term minimises as much as possible. If a single zero doesn't lie on the line, then the prediction will deviate significantly.

Imagine the complex plane as a piano and the zeta function as its musical chord. Then the critical line would be the central note. Considering the non-trivial zeroes as notes, if a single note were out of place, then the whole symphony of the prime numbers would be disturbed.

Let's talk about why this hypothesis is so important to prove. Prime numbers are frequently thought to be randomly distributed. If this hypothesis is true, then we can have perfect estimations of prime numbers. There are lots of other things riding on this conjecture. The proof of the *weak Goldbach Conjecture* [12], stating that any number greater than 5 can be expressed as the sum of three primes, requires the Riemann Hypothesis to be true. If the Hypothesis is true, then we will always find a prime between two consecutive cubes.

The Riemann Hypothesis also influences the realm of modern science. The value is very useful in physics especially in string theory and field theory. Famous physicist Freeman Dyson [13] connected the statistical pattern of zeta zeroes with the energy level distribution of atoms in quantum mechanics. This hypothesis appears in the study of quantum chaos, statistical mechanics, thermodynamics, black-body radiation and so on. It also influences modern technology like cryptography and spectroscopy.

In 1914, G.H. Hardy [3, 4] proved that infinitely many zeroes lie on the critical line but couldn't prove for all zeroes. In 1942, Atle Selberg [9] showed that a positive proportion of zeroes indeed lie on the critical line. However, he failed to show the percentage is 100. So far, trillions of zeroes have been found on the critical line using computers. Yet, neither a proof nor a counterexample has been found so far. Mathematicians are exploring numerous approaches that may open some ways to get to the solution. Due to its significant influence across mathematics and physics, this hypothesis is regarded as one of the most celebrated unsolved problem in mathematics. Famous mathematician David Hilbert in 1900 at the International Congress of Mathematicians in Paris presented a list of 23 unsolved problems as the challenge for the future generation. The Riemann Hypothesis was the 8th problem of the list. On 24 May 2000, the Clay Math institute listed the Riemann Hypothesis as one of the seven Millennium Prize Problems, each of whose prize amount is a million dollars.

Thousands of years ago, Euclid embarked on mankind's journey into the enigmatic world of primes by proving the infinitude of prime numbers. Centuries later, the Riemann hypothesis emerged as the North Star in the quest to discover the latent beauty of prime numbers. Whether solved by human genius or machine brilliance, the day this veil is lifted, it may not only answer the long-sought questions of number theory but also reveal a profound connection between mathematics and the natural world.

References

- [1] 3Blue1Brown, But what is the Riemann zeta function? Visualizing analytic continuation, *YouTube*, 2022.
- [2] J.E. Handmann, Portrait of Leonhard Euler (1707-1783), *Wikimedia Commons*, last accessed on 2025-12-20.
- [3] G.H. Hardy, Sur les Zéros de la Fonction $\zeta(s)$ de Riemann, *C. R. Acad. Sci. Paris* **158** (1914), 1012–1014.
- [4] G.H. Hardy and J.E. Littlewood, The zeros of Riemann’s zeta-function on the critical line, *Math. Z.* **10** (1921), 283–317.
- [5] W.L. Hosch and Britannica Editors, Riemann Hypothesis, *Encyclopedia Britannica*, last accessed on 2025-12-20.
- [6] Numberphile, Riemann Hypothesis - Numberphile, *YouTube*, 2013.
- [7] Quanta Magazine, The Riemann Hypothesis, Explained, *YouTube*, 2023.
- [8] B. Riemann, Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse, *Monatsberichte der Berliner Akademie*, 1859, in *Gesammelte Werke*, Teubner, Leipzig, 1892.
- [9] A. Selberg, On the zeros of Riemann’s zeta-function on the critical line, *Arch. Mat. Nat. B* **45** (1942), 101–114.
- [10] singingbanana, The Riemann Hypothesis, *YouTube*, 2021.
- [11] Unknown photographer, Portrait of Bernhard Riemann, *Wikimedia Commons*, last accessed on 2025-12-20.
- [12] Wikipedia, Goldbach’s weak conjecture, last accessed on 2025-12-20.
- [13] Wikipedia, Montgomery’s pair correlation conjecture, last accessed on 2025-12-20.
- [14] Wikipedia, Riemann hypothesis, last accessed on 2025-12-20.